Lesson notes · for you

How AI is attacked and defended: unit check

AI security · Upper secondary · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Filters that miss
- Only the tools it needs
- Poisoned data
- Check before you trust a model

Activity: Activity: Defend the class bot

1. Warm-up (3 min): Recall one defence from each lesson.
2. Plan (7 min): Design the defences for a made-up class bot.
3. Attack (5 min): A partner tries four attacks on paper.
4. Share (5 min): Which defence stopped each one?

Use made-up bots, messages and files only.

Discussion

- Which defence from this unit surprised you most?
- Where is a check in code better than a rule the model reads?
- How would you explain least privilege to a younger student?

Check yourself

- My most useful defence from this unit, and where I'd use it.

Try it this week

Pick the defence from this unit you'd use most (an action check, an allow-list, the neighbour check or a fingerprint) and explain it to someone at home with a real example this week.

Worksheet

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Name _____ Class _____ Date _____

1 **What is a prompt injection?**

- ◯ Orders hidden in text an AI app reads, hoping the model follows them
- ◯ A virus that drains the phone's battery
- ◯ A prompt so long that the model slows down

2 **Match each attack to the defence that holds against it.**

A reworded scam message	• • • •	Compare its fingerprint
An agent holding ten tools	• • • •	A neighbour check and majority labels
Flipped labels in training data	• • • •	An allow-list per task
A model file changed on the way	• • • •	A check on the action in code

3 **What does this program print?**

```
def can_send(text, asked, yes):
    return asked and yes

print(can_send("password", True, True))
print(can_send("Hello", False, True))
```

- ◯ True, then False
- ◯ False, then True
- ◯ False, then False

4 **A reading tool can never cause any harm.**

- ◯ True
- ◯ False

5 **Which of these are data poisoning? Choose all that apply.**

- ☐ A classmate flips labels in the class dataset as a joke
- ☐ Fake reviews posted to train a shop's model
- ☐ Training photos with a planted sticker that changes the answer
- ☐ Rain making one photo blurry